AI Engineering Applied Project

Personal MCP Server

ChatGPT Integration Project

Personal MCP server connecting ChatGPT to authenticated operational tools, with OAuth, backend integration and optional host administration.

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AUTHENTICATED ACCESS

Connects ChatGPT to remote tools through OAuth and JWT validation.

OPERATIONAL TOOLS

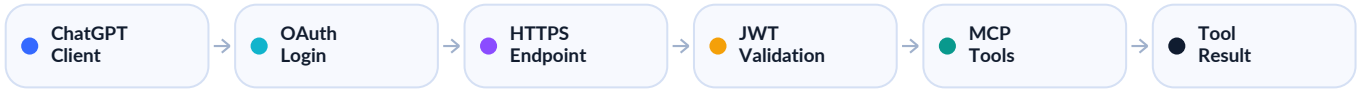
Exposes typed tools for health checks, runtime inspection and authorized actions.

REUSABLE INTEGRATION

Documents architecture, setup, diagnostics and an isolated browser extension.

Authenticated ChatGPT-to-Tools Workflow

The HTTPS endpoint, identity provider, MCP service and optional privileged bridge have separate roles.



Project Scope

- Remote MCP service exposed over HTTPS for personal ChatGPT use.
- OAuth and OpenID Connect integration through Keycloak.
- RS256 JWT validation with issuer, audience and identity checks.
- Typed tool discovery and authenticated operational requests.
- Optional Unix-socket bridge for explicitly authorized host operations.
- Separate Playwright browser integration present in deployed source.
- Public metadata checks and ChatGPT tool-catalog diagnostics.
- Reproducible setup guide with a minimal server example.

Technical Stack

Python · FastMCP · MCP · OAuth 2.0 · OIDC · Keycloak · PyJWT · Docker · HTTPS · Linux · Playwright · Backend APIs

Engineering Focus

Connecting a conversational interface to real operational capabilities with explicit authentication boundaries. The public service runs as a non-root container; privileged operations use a separate local bridge. Personal-use deployment and enterprise multiuser authorization remain distinct scopes.

Business Value

- Makes authorized operational tools accessible directly from ChatGPT.
- Reuses an existing identity provider and a typed tool interface.
- Separates conversational access from optional privileged execution.
- Provides a documented pattern for reproducing and diagnosing the integration.

Deployment & Validation

Deployed for personal use. Authenticated read checks and public 401 discovery verified; eight offline checks passed for the documented example. Browser execution, token renewal and multiuser authorization were outside this validation.

Project materials and engineering evidence reviewed - 03 Nov 2026.